MEDIUM · PROBLEM-SOLVING AND DATA ANALYSIS · ANSWER KEY

Two-variable data: Models and scatterplots

02

10 answers · Rationale-rich · Teach from doc

ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1

–

Answer not provided in source data — see rationale below.

Choice B is correct. Since the work is done at a constant rate, a linear model best models the situation. The number of shirts remaining is dependent on the length of time the inspector has worked; therefore, if the relationship were graphed, time would be the variable of the horizontal axis and the number of remaining shirts would be the variable of the vertical axis. Since the number of shirts decreases as the time worked increases, it follows that the slope of this graph is negative.

Choice A is incorrect and may result from incorrectly reasoning about the slope. Choices C and D are incorrect and may result from not identifying the constant rate of work as a characteristic of a linear model.

Q2 B

B. 7%

Choice B is correct. The charity with the greatest percent of total expenses spent on programs is represented by the highest point on the scatterplot; this is the point that has a vertical coordinate slightly less than halfway between 90 and 95 and a horizontal coordinate slightly less than halfway between 3,000 and 4,000. Thus, the charity represented by this point has a total income of about $3,400 million and spends about 92% of its total expenses on programs. The percent predicted by the line of best fit is the vertical coordinate of the point on the line of best fit with horizontal coordinate $3,400 million; this vertical coordinate is very slightly more than 85. Thus, the line of best fit predicts that the charity with the greatest percent of total expenses spent on programs will spend slightly more than 85% on programs. Therefore, the difference between the actual percent (92%) and the prediction (slightly more than 85%) is slightly less than 7%.

Choice A is incorrect. There is no charity represented in the scatterplot for which the difference between the actual percent of total expenses spent on programs and the percent predicted by the line of best fit is as much as 10%. Choices C and D are incorrect. These choices may result from misidentifying in the scatterplot the point that represents the charity with the greatest percent of total expenses spent on programs.

Q3 D

D. 2

Choice D is correct. A line in the *xy*-plane that passes through the points x_1, y_1 and x_2, y_2 has a slope of $\frac{y_2 - y_1}{x_2 - x_1}$. The line of best fit shown passes approximately through the points $1, 3.3$ and $7, 14.5$. It follows that the slope of this best fit line is approximately $\frac{14.5 - 3.3}{7 - 1}$, which is equivalent to $\frac{11.2}{6}$, or approximately 1.87. Therefore, of the given choices, 2 is closest to the slope of the line of best fit shown.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q4 C

C. There is a constant c such that if $0 < x < c$, then $2^x < 2x + 2$, but if $x > c$, then $2^x > 2x + 2$.

Choice C is correct. At $x = 0$, the value of 2^x is less than the value of $2x + 2$: $2^0 < 2(0) + 2$, which is equivalent to $1 < 2$. As the value of x increases, the value of 2^x remains less than the value of $2x + 2$ until $x = 3$, which is when the two values are equal: $2^3 = 2(3) + 2$, which is equivalent to $8 = 8$. Then, for $x > 3$, the value of 2^x is greater than the value of $2x + 2$. So there is a constant, 3, such that when $0 < x < 3$, then $2^x < 2x + 2$, but when $x > 3$, then $2^x > 2x + 2$.

Choice A is incorrect because $2^x > 2x + 2$ when $x > 3$. Choice B is incorrect because $2^x < 2x + 2$ when $0 < x < 3$. Choice D is incorrect because $2^x < 2x + 2$ when $0 < x < 3$ and $2^x > 2x + 2$ when $x > 3$.

Q5

0.5

Accept also: 1/2

The correct answer is $\frac{1}{2}$. For the graph shown, x represents time, in seconds, and y represents momentum, in newton-seconds. Therefore, the average rate of change, in newton-seconds per second, in the momentum of the object between two x-values is the difference in the corresponding y-values divided by the difference in the x-values. The graph shows that at $x = 2$, the corresponding y-value is 6. The graph also shows that at $x = 6$, the corresponding y-value is 8. It follows that the average rate of change, in newton-seconds per second, from $x = 2$ to $x = 6$ is $\frac{8 - 6}{6 - 2}$, which is equivalent to $\frac{2}{4}$, or $\frac{1}{2}$. Note that 1/2 and .5 are examples of ways to enter a correct answer.

Q6

A

A. $y = 3x + 0.8$

Choice A is correct. The data show a strong linear relationship between x and y. The line of best fit for a set of data is a linear equation that minimizes the distances from the data points to the line. An equation for the line of best fit can be written in slope-intercept form, $y = mx + b$, where m is the slope of the graph of the line and b is the y-coordinate of the y-intercept of the graph. Since, for the data shown, the y-values increase as the x-values increase, the slope of a line of best fit must be positive. The data shown lie almost in a line, so the slope can be roughly estimated using the formula for slope, $m = \frac{y_2 - y_1}{x_2 - x_1}$. The leftmost and rightmost data points have coordinates of about $(1,4)$ and $(8,26)$, so the slope is approximately $\frac{26-4}{8-1} = \frac{22}{7}$, which is a little greater than 3. Extension of the line to the left would intersect the y-axis at about $(0,1)$. Only choice A represents a line with a slope close to 3 and a y-intercept close to $(0,1)$.

Choice B is incorrect and may result from switching the slope and y-intercept. The line with a y-intercept of $(0,3)$ and a slope of 0.8 is farther from the data points than the line with a slope of 3 and a y-intercept of $(0,0.8)$. Choices C and D are incorrect. They represent lines with negative slopes, not positive slopes.

Q7

D

D.

x	1	2	3	4
y	6	12	24	48

Choice D is correct. The relationship between the values of x and their corresponding y-values is nonlinear if the rate of change between these pairs of values isn't constant. The table for choice D gives four pairs of values: $(1,6)$,$(2,12)$, $(3,24)$, and $(4,48)$. Finding the rate of change, or slope, between $(1,6)$ and $(2,12)$ by using the slope formula, $\frac{y_2-y_1}{x_2-x_1}$, yields $\frac{12-6}{2-1}$, or 6. Finding the rate of change between $(2,12)$ and $(3,24)$ yields $\frac{24-12}{3-2}$, or 12. Finding the rate of change between $(3,24)$ and $(4,48)$ yields $\frac{48-24}{4-3}$, or 24. Since the rate of change isn't constant for these pairs of values, this table shows a nonlinear relationship.

Choices A, B, and C are incorrect. The rate of change between the values of x and their corresponding y-values in each of these tables is constant, being 3, 4, and 5, respectively. Therefore, each of these tables shows a linear relationship.

Q8

6

The correct answer is 6. The line of best fit predicts a greater y-value than the actual y-value for any data point that's located below the line of best fit. For the scatterplot shown, 6 of the data points are below the line of best fit. Therefore, the line of best fit predicts a greater y-value than the actual y-value for 6 of the data points.

Q9 C

C. 6

Choice C is correct. Any data point that's located above the line of best fit has a y-value that's greater than the y-value predicted by the line of best fit. For the scatterplot shown, 6 of the data points are above the line of best fit. Therefore, 6 of the data points have an actual y-value that's greater than the y-value predicted by the line of best fit.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the number of data points that have an actual y-value that's less than the y-value predicted by the line of best fit.

Choice D is incorrect and may result from conceptual or calculation errors.

Q10 5

The correct answer is 5. For the graph shown, x represents time, in minutes, and y represents temperature, in degrees Celsius °C. Therefore, the average rate of change, in °C per minute, of the recorded temperature of the air in the chamber between two x-values is the difference in the corresponding y-values divided by the difference in the x-values. The graph shows that at $x = 5$, the corresponding y-value is 14. The graph also shows that at $x = 7$, the corresponding y-value is 24. It follows that the average rate of change, in °C per minute, from $x = 5$ to $x = 7$ is $\frac{24 - 14}{7 - 5}$, which is equivalent to $\frac{10}{2}$, or 5.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

Answers recovered from rationale for Q1 — the source CB data omitted a structured key for these items, so the answer slot shows —. The worked solution in the rationale below each entry contains the correct value.

Generated 2026-05-29 · cb-educator-question-bank